

CISCO CERTIFIED CYBEROPS ASSOCIATE (CBROPS 200-201)

Lab 17.1.7: Exploring DNS Traffic

Comprehensive Packet Analysis, Protocol Dissection, and Security Threat Evaluation

COURSE MODULE	PROTOCOL & PORT	ANALYSIS TOOL	WORKSTATION ENVIRONMENT
Module 17: Attacking What We Do	DNS / UDP & TCP Port 53	Wireshark & CLI (nslookup)	CyberOps Workstation VM

Objectives

- **Part 1: Capture DNS Traffic** — Configure packet capture environment, purge local resolver caches, and generate active DNS lookup traffic.
- **Part 2: Explore DNS Query Traffic** — Dissect protocol layers (Ethernet II, IPv4, UDP, DNS Application header), analyze bit flags, and correlate addressing.
- **Part 3: Explore DNS Response Traffic** — Inspect resolver response records, verify recursion capabilities, analyze CNAME aliasing, and correlate with CLI output.
- **Security Reflection & Threat Analysis** — Evaluate DNS reconnaissance risks, cleartext vulnerabilities, data exfiltration vectors, and SOC defensive controls.

Background / Scenario

The Domain Name System (DNS) is foundational to internet and enterprise operations, translating human-readable Fully Qualified Domain Names (FQDNs) into routable IP addresses. However, because standard DNS is unencrypted and connectionless (UDP port 53), it is frequently targeted by threat actors for network reconnaissance, cache poisoning (DNS spoofing), unauthorized exfiltration (DNS tunneling), and amplification Denial-of-Service (DoS) attacks.

In this lab, Wireshark is utilized as a protocol analyzer to capture and dissect live DNS transactions, evaluate header fields across the OSI stack, inspect recursive resolution mechanisms, and analyze the security implications of unencrypted DNS traffic within enterprise security operations.

Part 1: Capture DNS Traffic

Step 1: Download, Install, and Initialize Wireshark

Wireshark is launched on the CyberOps Workstation. When prompted during installation on physical environments, USBPcap is avoided to prevent driver conflicts on operational USB interfaces.

Step 2: Clear the DNS Cache and Generate Queries

To ensure Wireshark captures fresh over-the-network DNS queries rather than satisfying requests from local operating system cache, the resolver cache must be cleared prior to capture initialization:

```
# Windows DNS Cache Purge:
C:\> ipconfig /flushdns
Successfully flushed the DNS Resolver Cache.

# Linux (CyberOps Workstation VM - Systemd-Resolved):
analyst@secOps ~$ systemd-resolve --flush-caches
analyst@secOps ~$ sudo systemctl restart systemd-resolved.service

# macOS:
bash-3.2$ sudo killall -HUP mDNSResponder
```

Interactive `nslookup` session is executed to generate DNS traffic for `www.cisco.com`:

```
analyst@secOps ~$ nslookup
> www.cisco.com
Server:      192.168.1.1
Address:     192.168.1.1#53

Non-authoritative answer:
www.cisco.com canonical name = origin-www.cisco.com.akadns.net.
origin-www.cisco.com.akadns.net canonical name = e2867.dsca.akamaiedge.net.
Name:   e2867.dsca.akamaiedge.net
Address: 23.44.184.225
> exit
```

Part 2: Explore DNS Query Traffic

Applying display filter `udp.port == 53` or `dns` isolates DNS packets. Selecting the *Standard query 0x0001 A www.cisco.com* packet reveals the encapsulation stack:

STEP 2.D — ETHERNET II DATA LINK LAYER ANALYSIS

What are the source and destination MAC addresses? Which network interfaces are these MAC addresses associated with?

SOC ANALYST TECHNICAL FINDING:

- **Source MAC Address:** e.g., `08:00:27:da:28:83` (CyberOps Workstation Virtual NIC).
- **Destination MAC Address:** e.g., `50:c7:bf:dc:3a:41` (Default Gateway Router / Virtual NAT Gateway interface).
- **Interface Association:** The Source MAC belongs to the host PC's active physical/virtual Network Interface Card (NIC `enp0s3` / Ethernet adapter). The Destination MAC is associated with the local Default Gateway router's LAN interface, resolved via ARP prior to transmission.

STEP 2.E — IPV4 NETWORK LAYER ANALYSIS

What are the source and destination IP addresses? Which network interfaces are these IP addresses associated with?

SOC ANALYST TECHNICAL FINDING:

- **Source IPv4 Address:** `192.168.1.100` (or VM IP `10.0.2.15`), assigned to the local host's active network adapter.
- **Destination IPv4 Address:** `192.168.1.1` (or configured DNS resolver such as `209.165.200.225` / `8.8.8.8`).
- **Interface Association:** Source IP is bound to the local host operating system NIC. Destination IP is assigned to the upstream recursive DNS server or gateway DNS relay agent.

STEP 2.F — TRANSPORT LAYER ANALYSIS (UDP)

What are the source and destination ports? What is the default DNS port number?

SOC ANALYST TECHNICAL FINDING:

- **Source Port:** A dynamically assigned client ephemeral port (e.g., `53124` / `60481`, within range 49152–65535).
- **Destination Port:** `53` (Well-known port assigned to DNS).
- **Default DNS Port:** **UDP Port 53** is the universal default for standard queries and responses. (Note: TCP Port 53 is used for DNS Zone Transfers [AXFR/IXFR] and responses exceeding standard 512-byte UDP payload limits without EDNS0).

STEP 2.G — ADDRESS RESOLUTION & HOST BASELINE VERIFICATION

Compare the MAC and IP addresses in the Wireshark results to the IP and MAC addresses obtained from `ipconfig/arp`. What is your observation?

SOC ANALYST TECHNICAL FINDING:

The Source IP (`192.168.1.100`) and Source MAC (`08:00:27:da:28:83`) in the Wireshark frame perfectly match the Physical Address and IPv4 Address outputs from `ipconfig /all` (or Linux `ip addr`). Furthermore, the Destination MAC corresponds exactly to the ARP entry for the Default Gateway (`arp -a`), demonstrating that Layer 3 IP routing relies on Layer 2 MAC frame delivery across local broadcast domains.

PROTOCOL LAYER	FIELD INSPECTED	DNS QUERY VALUE	TECHNICAL SOC SIGNIFICANCE
Layer 2 (Data Link)	Ethernet II Frame	Src: Host NIC → Dst: Gateway MAC	Direct hop-to-hop frame transmission on the local LAN.
Layer 3 (Network)	IPv4 Header	Src: 192.168.1.100 → Dst: 192.168.1.1	Identifies initiating endpoint and configured DNS resolver.
Layer 4 (Transport)	User Datagram Protocol	Src Port: 53124 → Dst Port: 53	Connectionless transport minimizing lookup latency.
Layer 7 (Application)	DNS Query Header	Flags: <code>0x0100</code> (Recursion Desired = 1)	Client requests full recursive resolution from the server.

Part 3: Explore DNS Response Traffic

Inspecting the corresponding response frame (*Standard query response 0x0001 A www.cisco.com*) provides insight into server-side resolution:

STEP 3.A — ADDRESS & PORT SYMMETRY ANALYSIS

What are the source and destination MAC and IP addresses and port numbers? How do they compare to the addresses in the DNS query packets?

SOC ANALYST TECHNICAL FINDING:

- **Source MAC:** Gateway MAC | **Destination MAC:** Host PC Virtual NIC.
- **Source IP:** `192.168.1.1` (DNS Server) | **Destination IP:** `192.168.1.100` (Host PC).
- **Source Port:** `53` | **Destination Port:** `53124` (Client Ephemeral Port).
- **Comparison:** The Source and Destination addresses across Data Link (MAC), Network (IP), and Transport (UDP Port) layers are **precisely inverted (swapped)** compared to the query packet, confirming bidirectional communication symmetry.

STEP 3.C — DNS RECURSION FLAG INSPECTION

Can the DNS server do recursive queries?

SOC ANALYST TECHNICAL FINDING:

Yes. Within the DNS Response Flags (`0x8180`), the **Recursion Available (RA) bit is set to 1** (`1... .. = Recursion available: Server can do recursive queries`). This confirms the DNS server possesses recursive resolution capabilities and queried upstream authoritative servers on behalf of the client.

STEP 3.D — DNS RECORD CORRELATION WITH CLI OUTPUT

Observe the CNAME and A records in the Answers details. How do the results compare to nslookup results?

SOC ANALYST TECHNICAL FINDING:

The Wireshark DNS Answers section displays the complete resolution hierarchy:

1. `www.cisco.com` (CNAME) → `origin-www.cisco.com.akadns.net`
2. `origin-www.cisco.com.akadns.net` (CNAME) → `e2867.dsca.akamaiedge.net`
3. `e2867.dsca.akamaiedge.net` (A Record) → `23.44.184.225` (Akamai CDN Anycast IPv4 address)

These results match the output obtained from the interactive `nslookup` command identically, confirming that `nslookup` formats and prints the exact payload returned in the DNS Answer resource records.

Reflection & Threat Analysis Questions

REFLECTION QUESTION 1 — PASSIVE NETWORK RECONNAISSANCE

From the Wireshark results, what else can you learn about the network when you remove the filter?

SOC DEFENSIVE ASSESSMENT:

When display filters are removed, a packet capture exposes extensive operational intelligence across the broadcast domain:

- **Network Topology & Addressing Scheme:** Internal IPv4/IPv6 subnets, DHCP lease servers, Default Gateway IP/MAC addresses.
- **Broadcast / Multicast Protocols:** ARP discovery, LLNMR (Link-Local Multicast Name Resolution), mDNS (Bonjour), SSDP (UPnP devices), NBNS (NetBIOS).
- **Active Host Fingerprinting:** Operating system telemetry, hostnames, background software update services, connected IoT hardware.
- **Cleartext Application Data:** Unencrypted HTTP headers, FTP sessions, Telnet streams, plaintext NTP time sync requests, and unencrypted email (SMTP/IMAP).

REFLECTION QUESTION 2 — THREAT ACTOR WEAPONIZATION OF WIRESHARK

How can an attacker use Wireshark to compromise your network security?

THREAT VECTOR ANALYSIS:

Threat actors who obtain access to a network switchport (via ARP spoofing/port mirroring) or compromise an internal endpoint can utilize Wireshark for:

- **Credential & Session Sniffing:** Extracting plaintext passwords, API keys, session tokens, and authentication cookies transmitted over unencrypted protocols.
- **DNS Profiling & Behavioral Surveillance:** Monitoring DNS query requests to map all external domains, SaaS providers, and cloud services accessed by employees.
- **Man-in-the-Middle (MitM) Validation:** Confirming successful execution of ARP cache poisoning, rogue DHCP server deployment, or DNS spoofing attacks.
- **Vulnerability Reconnaissance:** Inspecting application banner exchanges (HTTP Server headers, SSH version strings, SMB dialects) to identify unpatched vulnerabilities for targeted exploitation.

CyberOps SOC Analyst Remediation & Mitigation Summary

- **DNS Encryption** Deploy **DNS over HTTPS (DoH - RFC 8484)** or **DNS over TLS (DoT - RFC 7858)** to prevent eavesdropping and metadata profiling on the local network.
- **Integrity Protection** Enforce **DNSSEC (DNS Security Extensions)** validation on recursive resolvers to cryptographically prevent DNS cache poisoning.
- **Tunneling Detection** Implement DNS inspection on Next-Gen Firewalls (NGFW) to detect and block DNS tunneling tools (e.g., Iodine, dnscat2) and Domain Generation Algorithms (DGA).
- **Layer 2 Hardening** Enable **Dynamic ARP Inspection (DAI)** and **DHCP Snooping** on access switches to prevent MitM packet sniffing.